

KSA Office IT Support & Infrastructure Management

Phase 05: Windows System Administration Implementation

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This phase documents the complete Windows system administration activities performed during the project. Rather than simply executing commands, each task was completed to simulate the responsibilities of a Windows System Administrator. Every command had a specific purpose, was verified through Windows tools or PowerShell, and was supported with screenshots to demonstrate successful implementation.

Module 01: System Information & Operating System Verification

Implementation Narrative:

This module was completed as a practical administration exercise. Each activity was performed, verified, and documented with screenshots to demonstrate both the execution process and the final outcome.

Commands and Purpose:

1. **winver** - Displayed the installed Windows edition and build number to confirm that the workstation met the project requirements before configuration began.
2. **msinfo32** - Opened the System Information console to review hardware components, BIOS, processor, installed memory, and operating system details. This provided a baseline inventory of the workstation.
3. **systeminfo** - Generated a detailed operating system report from the command line. The output was used to validate system configuration and document the environment.

Verification:

After completing the configuration, the results were validated using Windows graphical tools, PowerShell or Command Prompt output, and supporting screenshots captured during implementation.

Module 02: Local User and Group Administration

Implementation Narrative:

This module was completed as a practical administration exercise. Each activity was performed, verified, and documented with screenshots to demonstrate both the execution process and the outcome.

Commands and Purpose:

1. **net user** - Listed existing local accounts and verified that new departmental users were successfully created.
2. **net user <user> <password> /add** - Created department-specific user accounts to simulate employees from different business units.
3. **net user <user> /active:no** - Disabled a user account to demonstrate account management and security administration.
4. **net user <user> <newpassword>** - Reset an account password to simulate help-desk password recovery.
5. **net localgroup Administrators it.user /add** - Assigned administrative privileges to the IT account, demonstrating role-based access management.

Verification:

After completing the configuration, the results were validated using Windows graphical tools, PowerShell or Command Prompt output, and supporting screenshots captured during implementation.

Module 03: NTFS Permissions and Shared Folders

Implementation Narrative:

This module was completed as a practical administration exercise. Each activity was performed, verified, and documented with screenshots to demonstrate both the execution process and the final outcome.

Commands and Purpose:

1. **icacls C:\Afaq_Data** - Verified NTFS permissions after configuring access rights for departmental users.
2. **net share** - Displayed existing shared folders and confirmed that the project folder was published successfully.
3. **Computer Management → Shared Folders** - Verified the shared resources using the Windows management console.

Verification:

After completing the configuration, the results were validated using Windows graphical tools, PowerShell or Command Prompt output, and supporting screenshots captured during implementation.

Module 04: Windows Services Administration

Implementation Narrative:

This module was completed as a practical administration exercise. Each activity was performed, verified, and documented with screenshots to demonstrate both the execution process and the final outcome.

Commands and Purpose:

1. **sc query spooler** - Checked the operational status of the Print Spooler service to verify that it was running correctly.

2. **net start** - Displayed all running Windows services to confirm core operating system services were available.
3. **Get-Service | Export-Csv** - Exported service information into a report for documentation and future analysis.

Verification:

After completing the configuration, the results were validated using Windows graphical tools, PowerShell or Command Prompt output, and supporting screenshots captured during implementation.

Module 05: Device and Driver Management

Implementation Narrative:

This module was completed as a practical administration exercise. Each activity was performed, verified, and documented with screenshots to demonstrate both the execution process and the final outcome.

Commands and Purpose:

1. **driverquery /v** - Generated a complete inventory of installed drivers and their operational status.
2. **wmic computersystem get manufacturer,model** - Verified the workstation manufacturer and model.
3. **wmic cpu get name** - Retrieved processor information.
4. **wmic diskdrive get model,size** - Documented installed storage devices and capacities.

Verification:

After completing the configuration, the results were validated using Windows graphical tools, PowerShell or Command Prompt output, and supporting screenshots captured during implementation.

Module 06: Disk and Storage Administration

Implementation Narrative:

This module was completed as a practical administration exercise. Each activity was performed, verified, and documented with screenshots to demonstrate both the execution process and the final outcome.

Commands and Purpose:

1. **Disk Management** - Reviewed disk partitions, drive letters and storage configuration.
2. **Get-Volume | Export-Csv** - Exported storage information into an Excel-compatible report for documentation.

Verification:

After completing the configuration, the results were validated using Windows graphical tools, PowerShell or Command Prompt output, and supporting screenshots captured during implementation.

Module 07: Event Viewer and System Monitoring

Implementation Narrative:

This module was completed as a practical administration exercise. Each activity was performed, verified, and documented with screenshots to demonstrate both the execution process and the final outcome.

Commands and Purpose:

1. **Event Viewer** - Reviewed Windows logs to identify warnings and operational events.
2. **Get-Process | Export-Csv** - Captured all running processes for performance documentation.
3. **(Get-CimInstance Win32_OperatingSystem).LastBootUpTime** - Verified the system uptime to assess workstation availability.

Verification:

After completing the configuration, the results were validated using Windows graphical tools, PowerShell or Command Prompt output, and supporting screenshots captured during implementation.

Module 08: Windows Security Administration

Implementation Narrative:

This module was completed as a practical administration exercise. Each activity was performed, verified, and documented with screenshots to demonstrate both the execution process and the final outcome.

Commands and Purpose:

1. **Get-MpComputerStatus** - Verified Microsoft Defender health, signatures, and protection status.
2. **Get-NetFirewallRule | Export-Csv** - Exported Windows Firewall rules to document the configured security policy.

Verification:

After completing the configuration, the results were validated using Windows graphical tools, PowerShell or Command Prompt output, and supporting screenshots captured during implementation.

Module 09: Backup, Recovery and System Maintenance

Implementation Narrative:

This module was completed as a practical administration exercise. Each activity was performed, verified, and documented with screenshots to demonstrate both the execution process and the final outcome.

Commands and Purpose:

1. **DISM /Online /Cleanup-Image /CheckHealth** - Performed a quick assessment of the Windows component store for corruption.

2. **DISM /Online /Cleanup-Image /ScanHealth** - Executed a detailed scan to determine whether component corruption existed and whether repair would be required.
3. **System Protection** - Reviewed restore point configuration and recovery settings to verify disaster recovery readiness.

Verification:

After completing the configuration, the results were validated using Windows graphical tools, PowerShell or Command Prompt output, and supporting screenshots captured during implementation.

Phase Conclusion:

By the end of this phase, the workstation had been administered using industry-standard Windows utilities and command-line tools. The implementation demonstrated practical skills in system verification, account administration, permissions management, services, device management, storage, monitoring, security, and maintenance. Each module concluded with verification to ensure that the expected configuration had been successfully applied.